

# Supporting Information

## Uncertainty Quantification and Calibration in Template-Based Retrosynthesis Models: A Cross-Architecture Study of LocalRetro and MHNreact

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The following is available free of charge: single-architecture detail, distribution plots, ROC curves, and secondary tables that support the main-text AUROC and ECE numbers without adding new claims (this document, following section). Nothing here contradicts or extends the main text; it exists for readers who want to verify a specific number’s provenance or see a full single-model breakdown.

### S1 Baseline UQ Detail

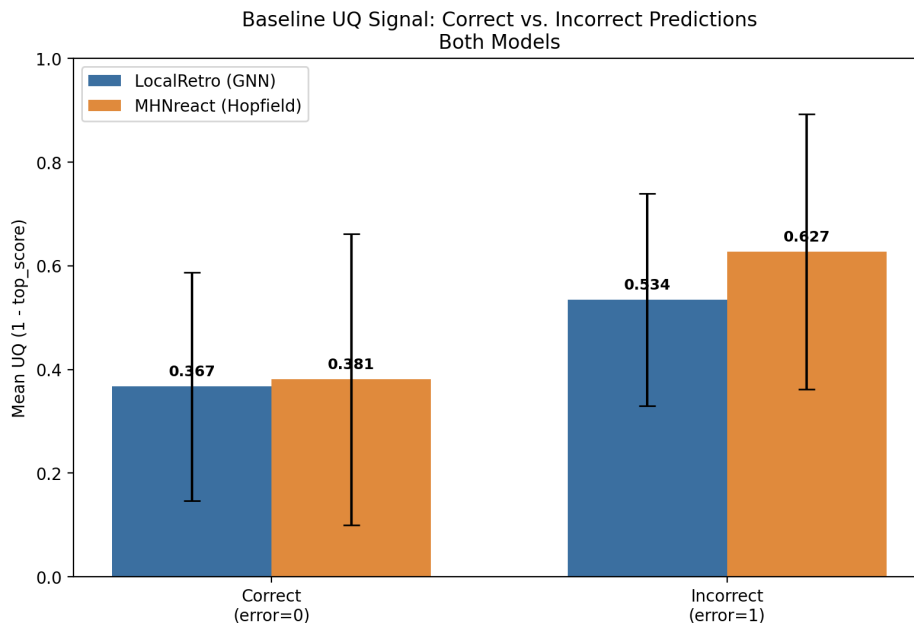


Figure S1: Mean baseline UQ signal ( $1 - \text{top\_score}$ ), correct vs. incorrect predictions, both models. Error bars show  $\pm 1$  SD. Supports the means reported in Table 1.

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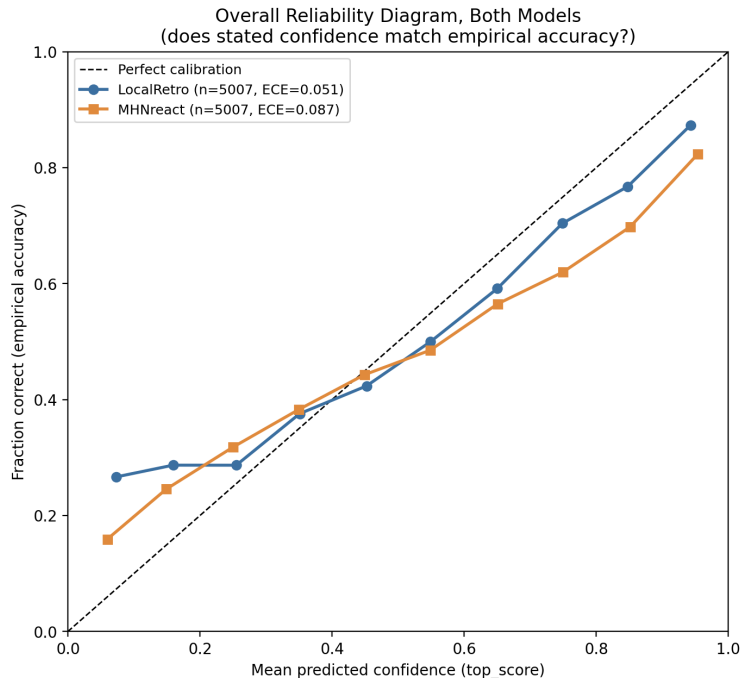


Figure S2: Overall reliability diagram, both models. LocalRetro ( $\text{ECE} = 0.051$ ) is better calibrated on average than MHNreact ( $\text{ECE} = 0.087$ ); both curves sit close to the diagonal in the mid-confidence range and diverge somewhat at the extremes.

## S2 Cross-Model Comparison Detail

The reliability-by-tier comparison is now presented in the main text alongside Table 1 (Figure 1).

## S3 Single-Architecture OOD Calibration Detail

### S3.1 LocalRetro

| Frequency tier     | n     | ECE (EW) | ECE (EM) | Bias floor | ECE (debiased) | MCE   |
|--------------------|-------|----------|----------|------------|----------------|-------|
| Very rare (1–5)    | 100   | 0.392    | 0.379    | 0.101      | 0.291          | 0.712 |
| Rare (6–20)        | 176   | 0.278    | 0.281    | 0.077      | 0.200          | 0.610 |
| Common (21–100)    | 486   | 0.160    | 0.157    | 0.047      | 0.113          | 0.300 |
| Very common (100+) | 4,245 | 0.038    | 0.042    | 0.016      | 0.024          | 0.214 |
| Overall            | 5,007 | 0.051    | 0.052    | 0.014      | 0.037          | 0.193 |

Table S1: LocalRetro: ECE by training-template frequency tier. EW = equal-width bins; EM = equal-mass (quantile) bins; Bias floor = expected ECE under perfect calibration at this  $n$ ; Debiased = raw EW minus bias floor.

Cross-Architecture Comparison: Rank-Based Vocabulary-Matched Tiers  
(n=500/1152/1652/1703 per tier, same for both models; MHNreact retrained with verified early stopping)

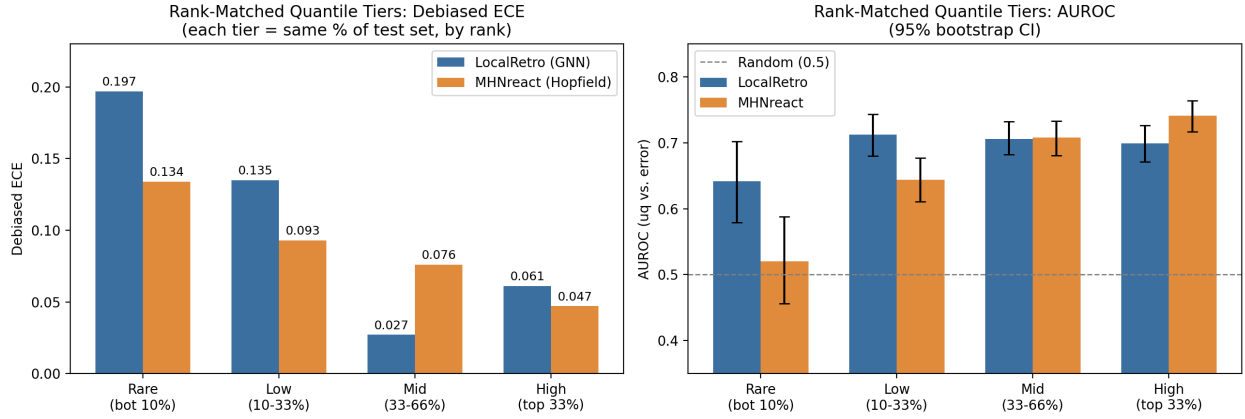


Figure S3: Rank-based vocabulary-matched tier comparison (see main-text Table 2 for the numbers). *Left:* Debiased ECE by tier. Both models have highest ECE in the rare tier, but their profiles differ: LocalRetro shows a sharp drop from rare to mid (0.197 to 0.027), while MHNreact declines smoothly and monotonically (0.134 to 0.047). *Right:* AUROC by tier with 95% bootstrap CIs; both models maintain discrimination above random across all tiers.

| Tier               | n     | Acc   | ECE (EW) | ECE (EM) | Bias  | ECE (deb) | AUROC                |
|--------------------|-------|-------|----------|----------|-------|-----------|----------------------|
| Very rare (1–5)    | 1,835 | 0.284 | 0.129    | 0.129    | 0.022 | 0.107     | 0.634 [0.608, 0.661] |
| Rare (6–20)        | 785   | 0.454 | 0.081    | 0.090    | 0.036 | 0.045     | 0.722 [0.689, 0.757] |
| Common (21–100)    | 1,097 | 0.540 | 0.097    | 0.098    | 0.030 | 0.068     | 0.720 [0.688, 0.750] |
| Very common (100+) | 1,290 | 0.674 | 0.065    | 0.068    | 0.027 | 0.038     | 0.741 [0.714, 0.770] |
| Overall            | 5,007 | 0.467 | 0.087    | 0.090    | 0.014 | 0.073     | 0.738 [0.724, 0.752] |

Table S2: MHNreact calibration by absolute-count frequency tier, retrained with verified early stopping (secondary; see Table 2 for the primary vocabulary-matched comparison). The debiased ECE series (0.038  $\rightarrow$  0.068  $\rightarrow$  0.045  $\rightarrow$  0.107 from most to least common) is *not* monotonic on this binning scheme, though the rank-based tiers are cleanly monotonic; this likely reflects tier-composition sensitivity rather than a true non-monotonic calibration trend.

### S3.2 MHNreact

## S4 Temperature Scaling Detail

## S5 Held-Out-Class Confidence Detail

## S6 Perturbation UQ Detail

### S6.1 MC Dropout

### S6.2 Deep Ensemble

## S7 Mondrian Conformal Detail

## S8 Risk-Coverage Curves

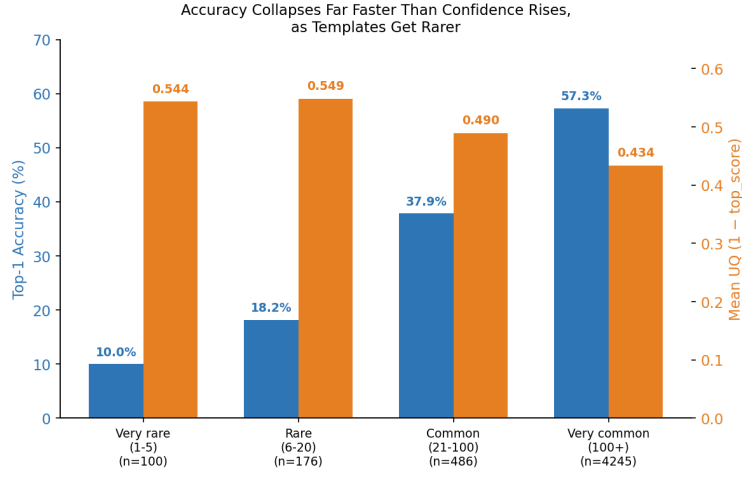


Figure S4: LocalRetro: top 1 accuracy (left axis) and mean UQ (right axis) across four training template frequency tiers. Confidence rises only modestly even as accuracy collapses.

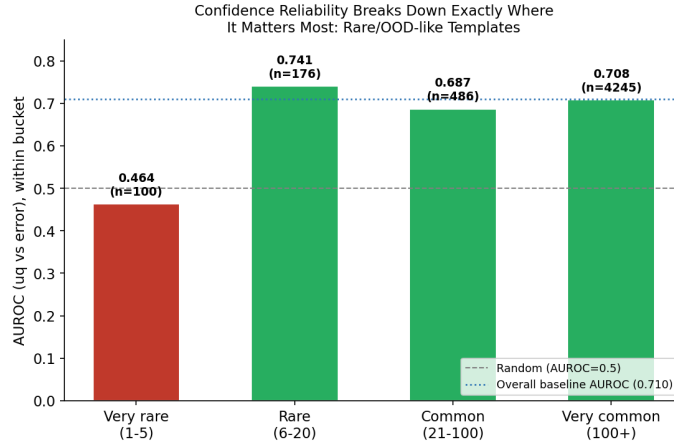


Figure S5: LocalRetro: AUROC (uq vs. error) computed separately within each training frequency tier, with bootstrap 95% CIs.

| Signal                      | AUROC  | 95% CI         | Spearman $\rho$ |
|-----------------------------|--------|----------------|-----------------|
| 1 – top_score (baseline)    | 0.7104 | [0.696, 0.724] | 0.3637          |
| 1 – mean_top_score (MC avg) | 0.7063 | [0.692, 0.720] | 0.3566          |
| std_top_score               | 0.5446 | [0.530, 0.560] | 0.0771          |
| 1 – agreement_rate          | 0.6420 | [0.628, 0.657] | 0.2698          |
| predictive_entropy          | 0.5922 | [0.576, 0.608] | 0.1594          |

Table S3: LocalRetro: full signal comparison with bootstrap 95% CIs (2,000 resamples).

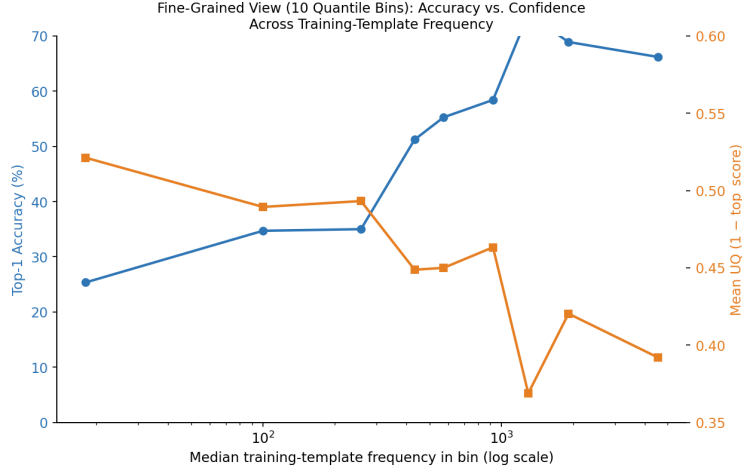


Figure S6: LocalRetro: fine grained (10 quantile bins) accuracy and mean confidence vs. median training template frequency (log x axis).

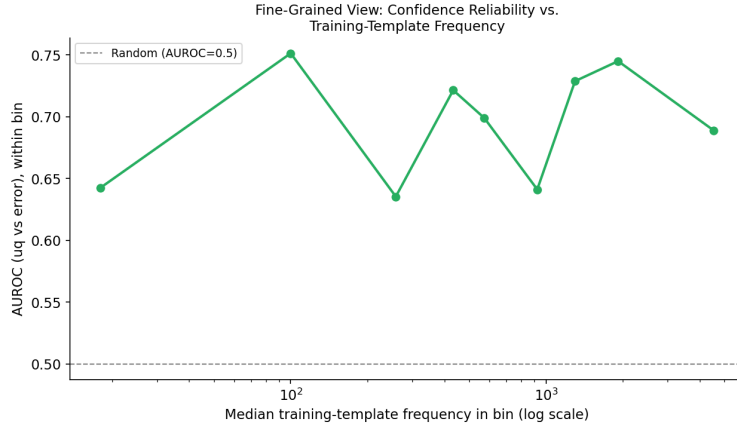


Figure S7: LocalRetro: fine grained within bin AUROC vs. median training template frequency. Reliability stays broadly stable (0.64 to 0.75) outside the most extreme rarity tail.

| Signal                      | AUROC  | 95% CI         | Spearman $\rho$ |
|-----------------------------|--------|----------------|-----------------|
| 1 - top_score (baseline)    | 0.7376 | [0.724, 0.752] | 0.4107          |
| 1 - mean_top_score (MC avg) | 0.7240 | [0.709, 0.738] | 0.3872          |
| std_top_score               | 0.3295 | [0.315, 0.345] | -0.2947         |
| 1 - agreement_rate          | 0.7103 | [0.697, 0.724] | 0.3698          |
| predictive_entropy          | 0.6942 | [0.679, 0.709] | 0.3356          |

Table S4: MHNreact: MC dropout signal comparison, all  $p < 10^{-100}$ .

| Signal                         | Overall AUROC        | Very-rare AUROC | Very-rare Bonf. CI |
|--------------------------------|----------------------|-----------------|--------------------|
| Baseline (1 - top_score)       | 0.710 [0.696, 0.725] | 0.464           | [0.201, 0.761]     |
| Ensemble mean (1 - $\bar{p}$ ) | 0.727 [0.713, 0.741] | 0.491           | [0.234, 0.766]     |
| Ensemble std (disagreement)    | 0.549 [0.534, 0.565] | 0.601           | [0.361, 0.812]     |

Table S5: LocalRetro deep ensemble results (4 seeds). All very-rare-tier CIs use Bonferroni correction across 3 signals (n=100).

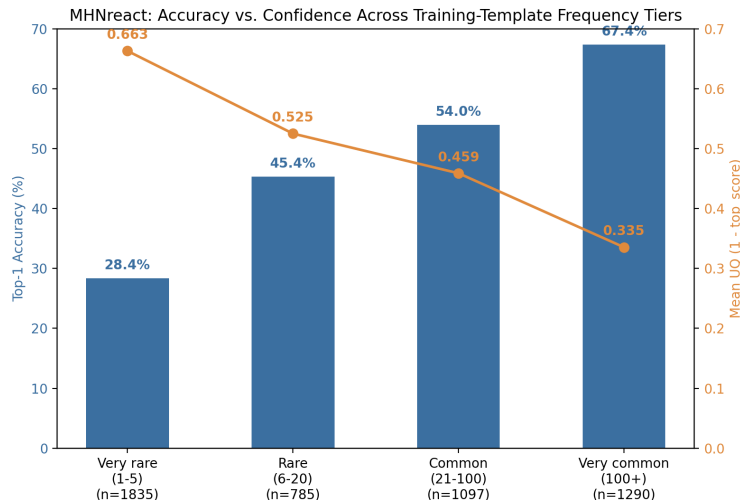


Figure S8: MHNreact: top-1 accuracy (left axis) and mean UQ (right axis) across four training-template frequency tiers. Accuracy rises from 28.4% to 67.4%, a 39-point spread, versus LocalRetro’s 47.3-point collapse from a much smaller very-rare base rate.

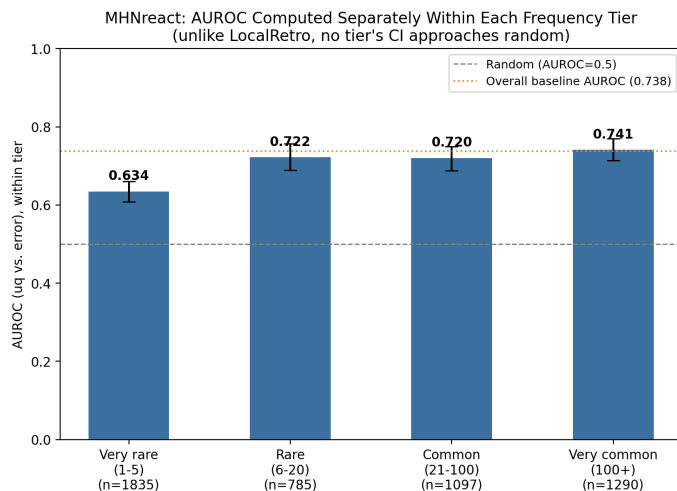


Figure S9: MHNreact: AUROC (uq vs. error) computed separately within each frequency tier, with bootstrap 95% CIs.

| Signal                               | Overall AUROC        | Very-rare AUROC | Very-rare Bonf. CI |
|--------------------------------------|----------------------|-----------------|--------------------|
| Baseline ( $1 - \text{top\_score}$ ) | 0.738 [0.724, 0.752] | 0.634           | [0.602, 0.665]     |
| Ensemble mean ( $1 - \bar{p}$ )      | 0.738 [0.724, 0.752] | 0.645           | [0.614, 0.676]     |
| Ensemble std (disagreement)          | 0.499 [0.484, 0.515] | 0.427           | [0.394, 0.461]     |

Table S6: MHNreact deep ensemble results (4 seeds: 1–4, seed 0 is the baseline).

| Ensemble           | Very-rare AUROC | Bonferroni CI  |
|--------------------|-----------------|----------------|
| First (seeds 1–4)  | 0.4270          | [0.394, 0.461] |
| Second (seeds 5–8) | 0.4277          | [0.394, 0.460] |

Table S7: Ensemble-disagreement inversion, two independently trained 4-seed ensembles. Both land within 0.001 AUROC of each other with essentially identical CIs, despite sharing no random initializations.

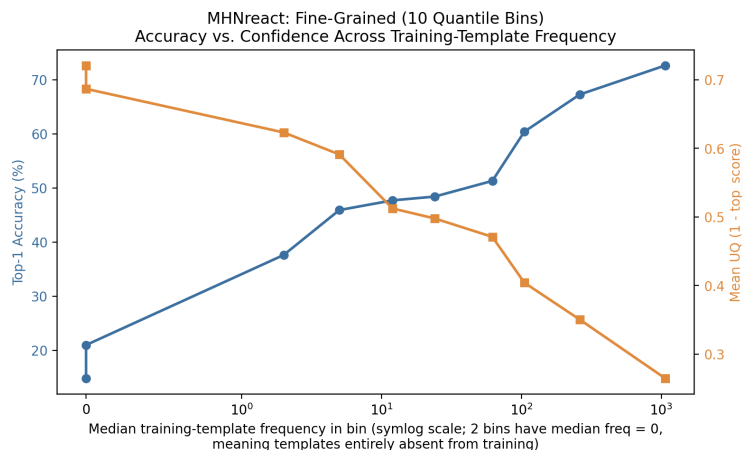


Figure S10: MHNreact: fine-grained (10 quantile bins) accuracy and mean confidence vs. median training-template frequency (symlog x-axis, since 2 bins have median frequency 0).

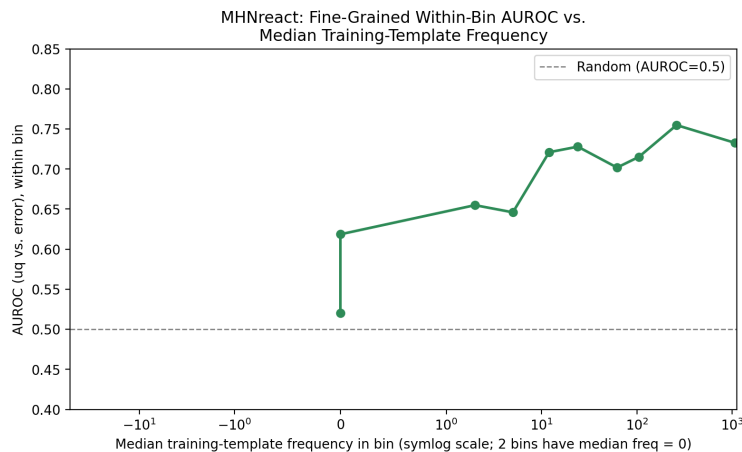


Figure S11: MHNreact: fine-grained within-bin AUROC vs. median training-template frequency. AUROC rises from 0.52 in the zero-frequency bin to a plateau around 0.70–0.75 for established templates, a smoother, more monotonic climb than LocalRetro’s fluctuating 0.64–0.75 range.

| Tier               | n (test half) | Global: kept / err | Mondrian: kept / err | Base error rate |
|--------------------|---------------|--------------------|----------------------|-----------------|
| Very rare (1–5)    | 936           | 4.4% / 63.4%       | 0.0% (abstains)      | 70.8%           |
| Rare (6–20)        | 383           | 17.5% / 31.3%      | 22.2% / 30.6%        | 51.2%           |
| Common (21–100)    | 543           | 20.4% / 18.9%      | 31.9% / 22.0%        | 45.5%           |
| Very common (100+) | 642           | 31.3% / 13.4%      | 54.8% / 16.8%        | 30.2%           |

Table S8: MHNreact Mondrian conformal results (naive threshold), target error rate 20%. “Global” and “Mondrian” columns show fraction kept (coverage) / error rate among kept predictions. See main-text Table 5 for the Hoeffding-UCB-corrected version of this same analysis.

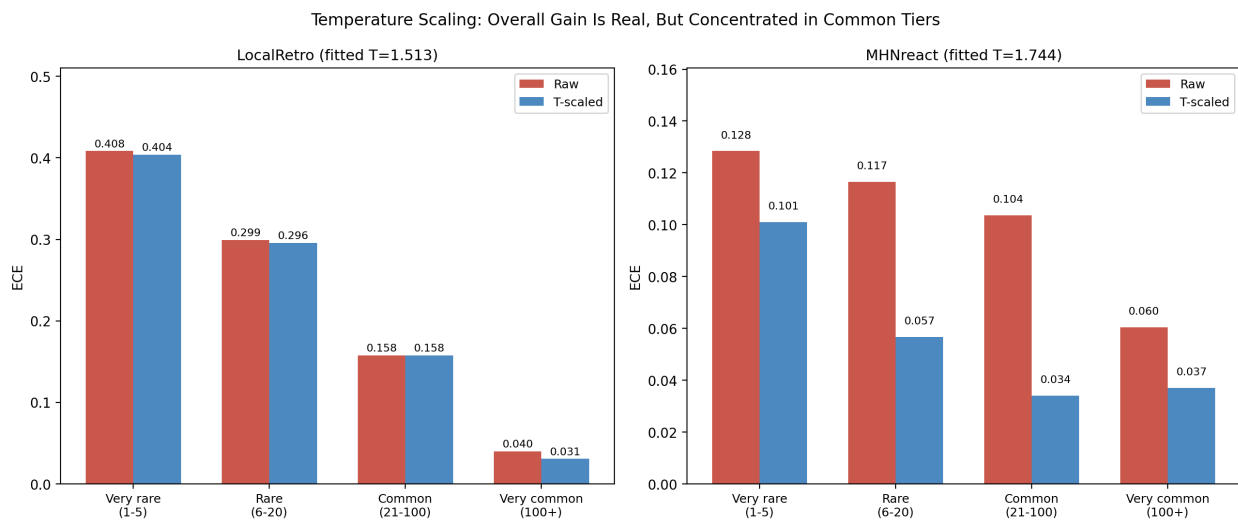


Figure S12: Temperature scaling, raw vs. T-scaled ECE by tier, both models (see main-text Table 3 for the numbers). LocalRetro’s rare tiers barely move and the common tier is unchanged; nearly all of its overall gain comes from the very-common tier. MHNreact improves more evenly across tiers, consistent with its uniformly-miscalibrated signature rather than a sharp collapse.

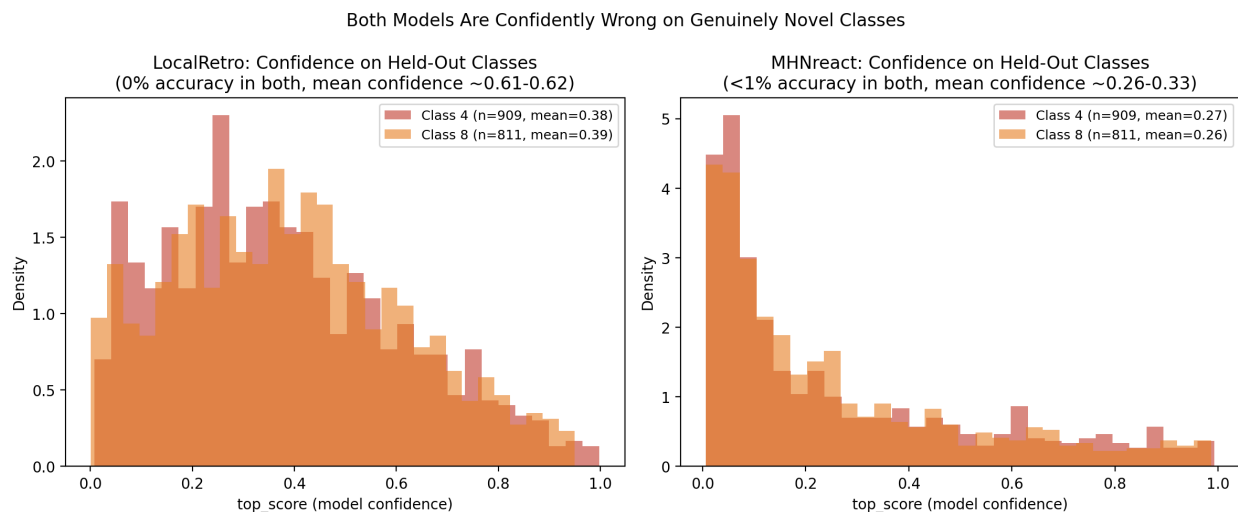


Figure S13: Confidence distribution on the held-out (OOD) test sets, both models (see main-text Table 4 for the ECE values). Despite 0% (LocalRetro) or near-0% (MHNreact) actual accuracy, mean confidence stays well above zero in every case (LocalRetro: 0.376–0.390; MHNreact: 0.259–0.269): both models remain confidently wrong on genuinely novel reaction classes rather than correctly signaling low confidence, the calibration failure mode most relevant to deployment.

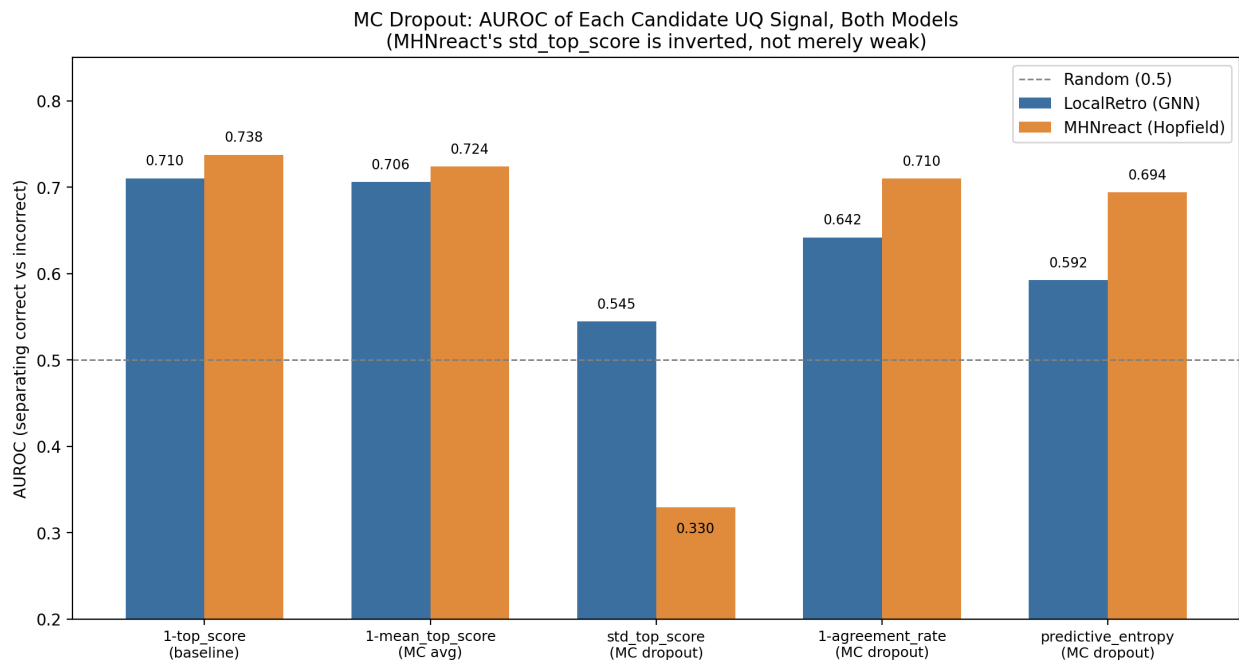


Figure S14: AUROC of each candidate MC dropout signal, both models. MHNreact's std\_top\_score is not merely weak like LocalRetro's; it is inverted, falling well below the random-chance line.

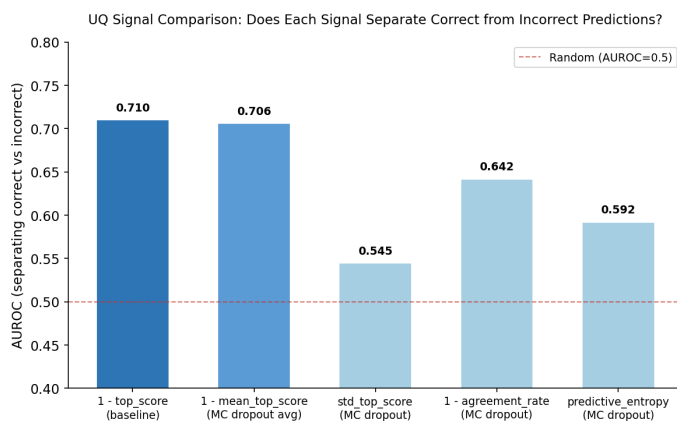


Figure S15: LocalRetro: AUROC of each candidate UQ signal in separating correct from incorrect predictions. Dashed line = random (AUROC = 0.5).

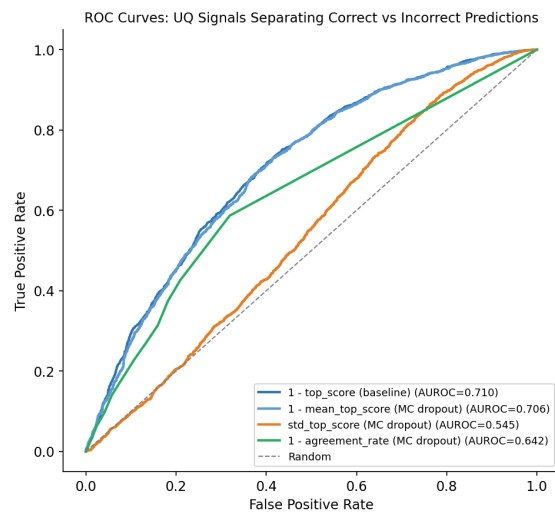


Figure S16: LocalRetro: ROC curves for all evaluated UQ signals.

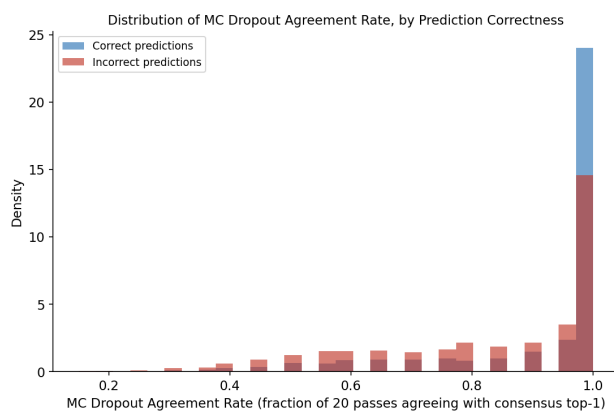


Figure S17: LocalRetro: distribution of MC dropout agreement\_rate, by correctness.

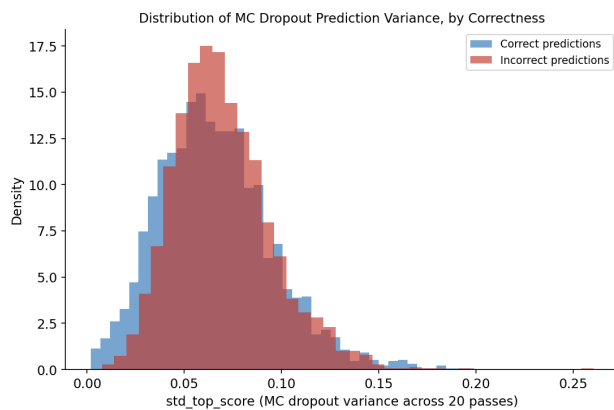


Figure S18: LocalRetro: distribution of MC dropout prediction variance (std\_top\_score), by correctness.

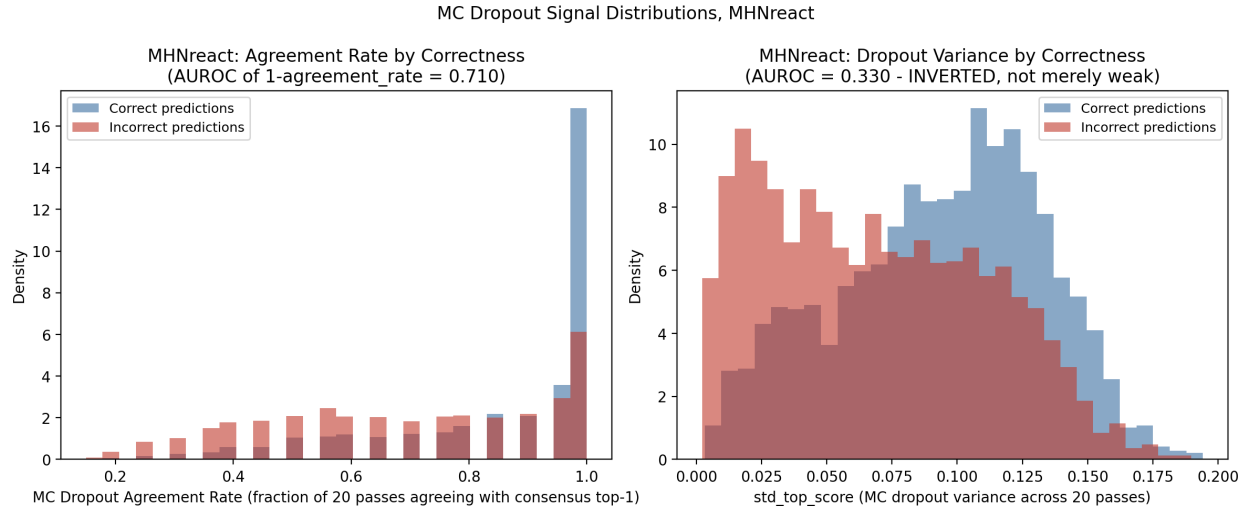


Figure S19: MHNreact: MC dropout signal distributions by correctness. *Left*: agreement rate. *Right*: dropout variance, showing the inversion visually.

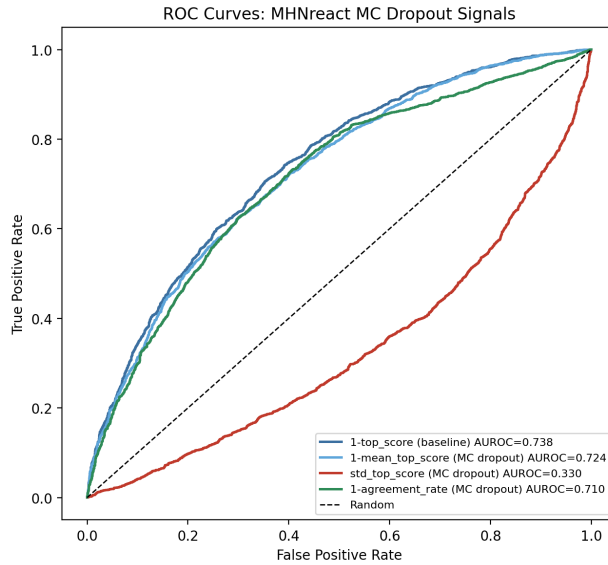


Figure S20: MHNreact: ROC curves for MC dropout signals. The `std_top_score` curve sits visibly below the diagonal.

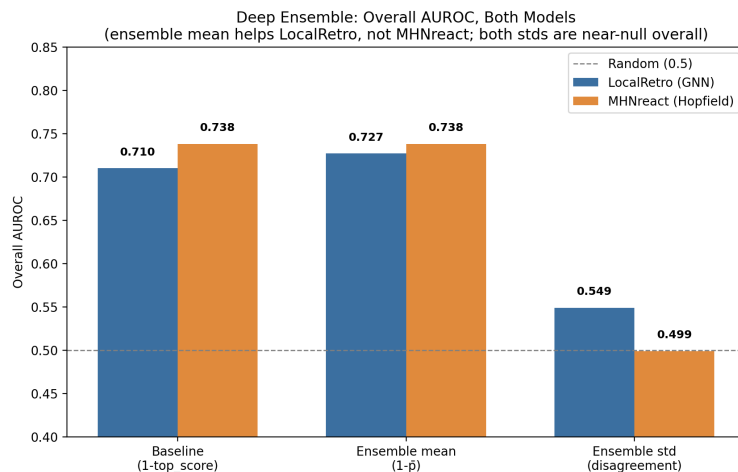


Figure S21: Deep ensemble overall (non-tiered) AUROC, both models. Ensemble mean helps LocalRetro but not MHNreact; both ensemble-std signals hover near random overall, which is why the very-rare-tier breakdown below is the figure that carries the argument.

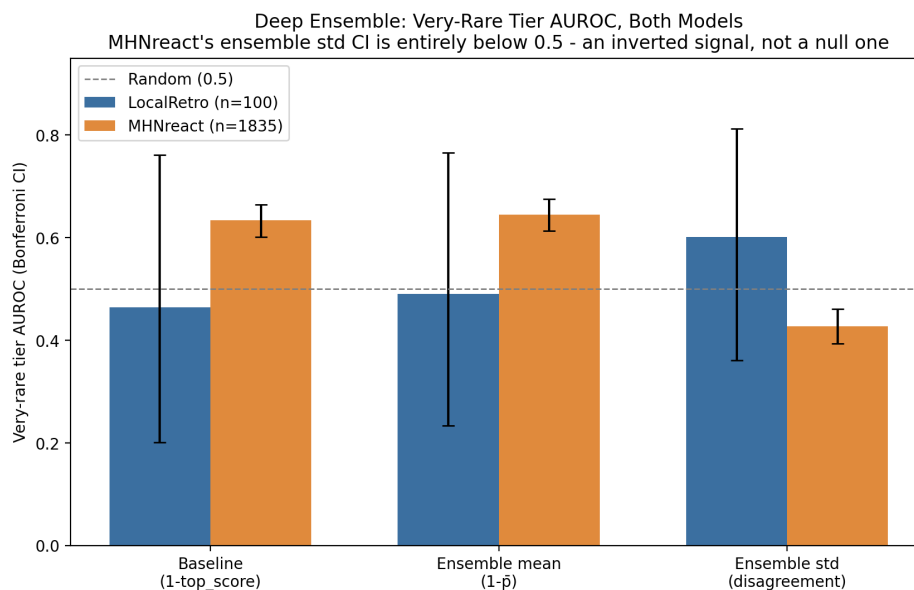


Figure S22: Very-rare tier AUROC with Bonferroni CIs, both models. LocalRetro's ensemble std CI spans 0.5 (a clean null). MHNreact's ensemble std CI sits entirely below 0.5, a statistically real inversion, not a wider, less-certain null.

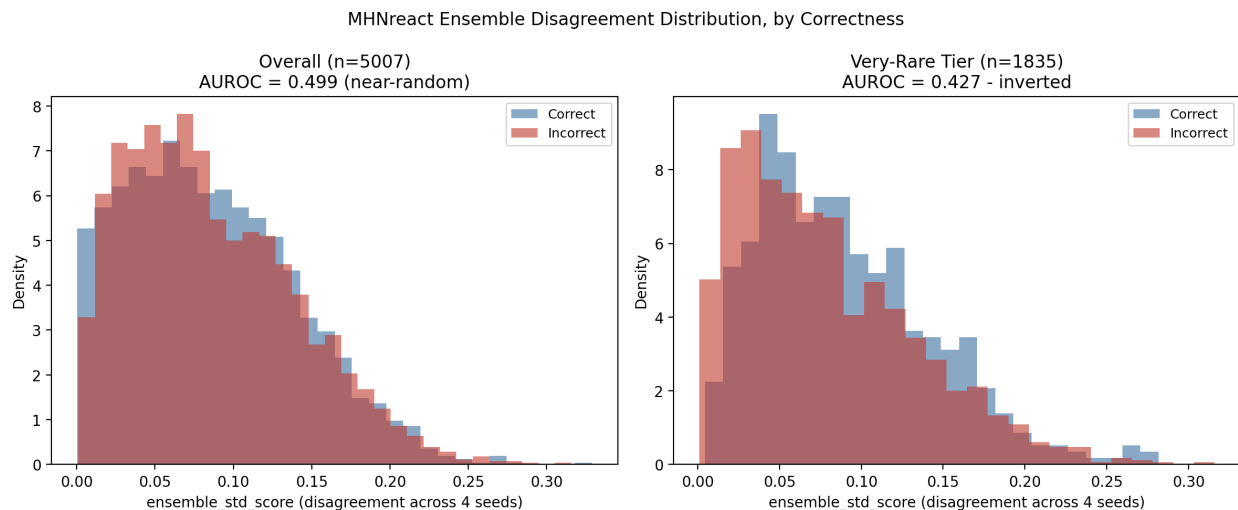


Figure S23: MHNreact ensemble disagreement distribution by correctness, overall vs. very-rare tier. The rightward shift of the “correct” distribution relative to “incorrect” in the very-rare panel is the visual signature of the inversion.

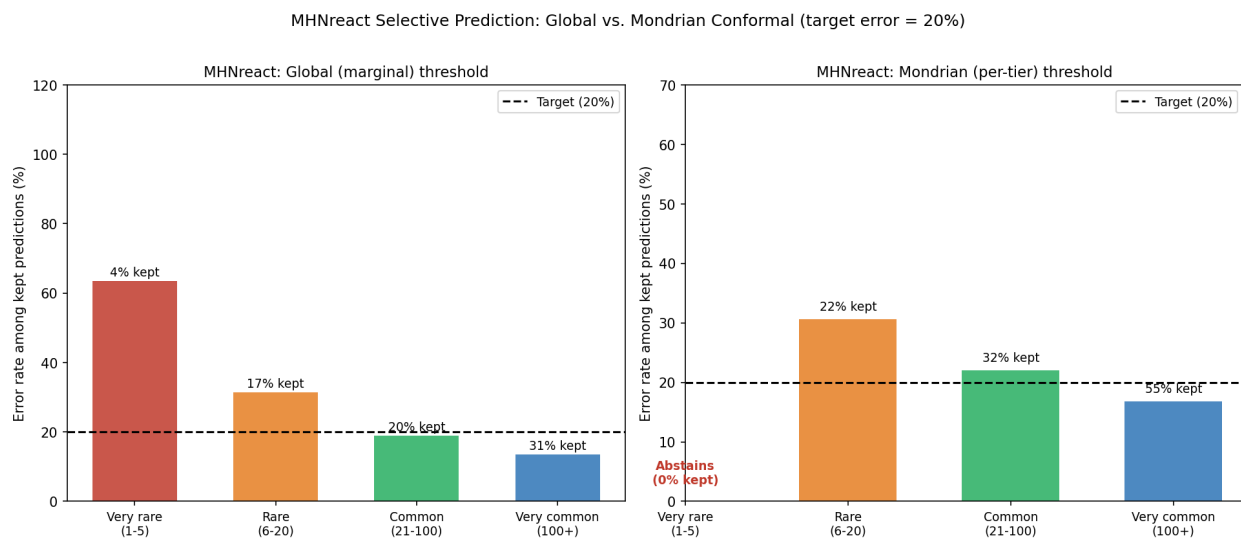


Figure S24: MHNreact selective prediction under a 20% target error rate, same harness as LocalRetro (Figure 5). *Left*: global threshold. *Right*: Mondrian per-tier thresholds.

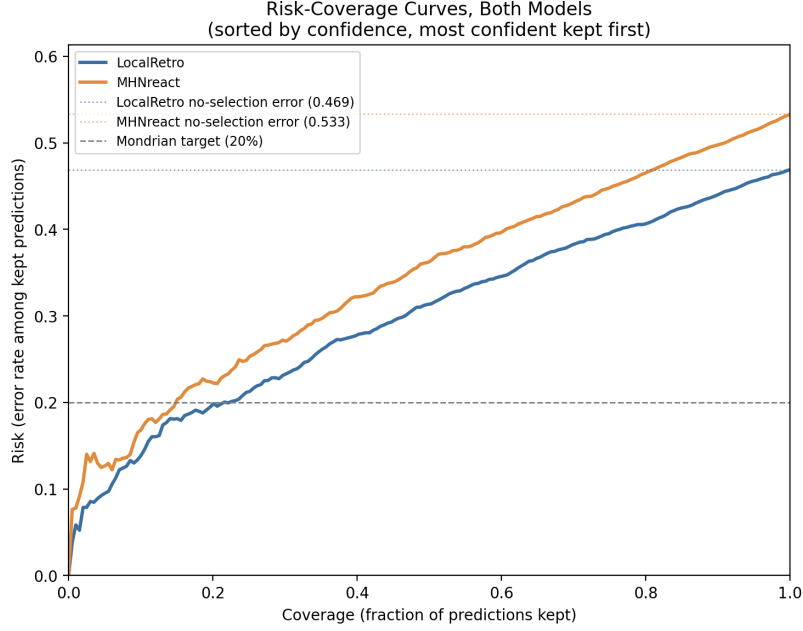


Figure S25: Overall risk-coverage curves, both models (sorted by confidence, most-confident-first, no tier stratification).

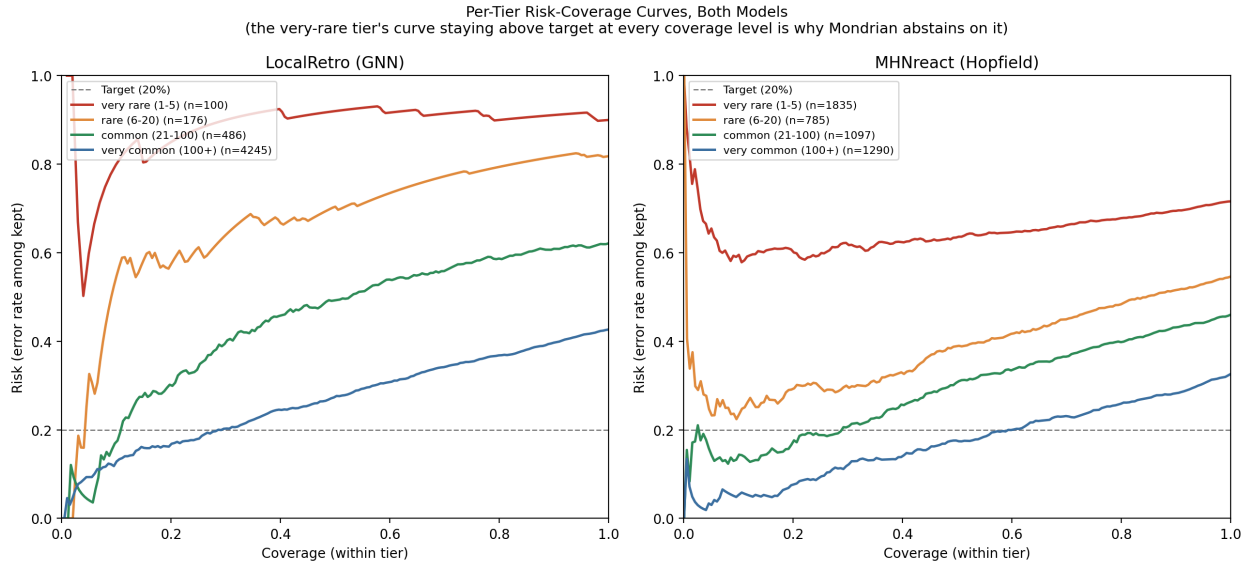


Figure S26: Per-tier risk-coverage curves, both models. LocalRetro's very-rare-tier curve (red, left panel) never drops to the 20% target line even at very low coverage. MHNreact's tiers (right panel) are more tightly clustered and each eventually crosses the target at some coverage level, though the rare tier crosses it only at low coverage, consistent with the finite-sample slack documented in Section 8.